

MACRO-II

```
#include <stdio.h>
```

```
#include <string.h>
```

```
char mdt[50][50] = {  
    "MOVER (P,3) (P,1)",  
    "ADD (P,3) (P,2)",  
    "MOVEM (P,3) (P,1)",  
    "MEND"  
};
```

```
char pntab[5][15] = {"", "&MEM_VAL", "&INCR_VAL", "&REG"};  
char actual_params[5][15]; // ANTAB
```

```
void main() {  
    char call[] = "INCR A, B, AREG";  
    char *token;  
    int i;  
    clrscr();  
    // Parse macro call and populate ANTAB  
    token = strtok(call, " ,");  
    token = strtok(NULL, " ,"); // Skip macro name  
  
    i = 0;  
    while (token != NULL) {  
        strcpy(actual_params[++i], token);  
        token = strtok(NULL, " ,");  
    }
```

```
}
```

```
// Print ANTAB
```

```
printf("\tANTAB (Actual Name Table)\n");
```

```
printf("-----\n");
```

```
printf("Index | Formal Parameter | Actual Arg\n");
```

```
printf("-----\n");
```

```
for (i = 1; i <= 3; i++) {
```

```
printf(" %d | %-15s | %s\n", i, pntab[i], actual_params[i]);
```

```
}
```

```
printf("-----\n\n");
```

```
// Expand macro
```

```
printf("Expanded Code:\n-----\n");
```

```
for (i = 0; strcmp(mdt[i], "MEND") != 0; i++) {
```

```
char line[80] = "+ ", temp[80], *p;
```

```
strcpy(temp, mdt[i]);
```

```
p = strtok(temp, " \n");
```

```
while (p != NULL) {
```

```
    if (strncmp(p, "(P,", 3) == 0) {
```

```
        int index = p[3] - '0';
```

```
        strcat(line, actual_params[index]);
```

```
    } else {
```

```
        strcat(line, p);
```

```
    }
```

```
    strcat(line, " ");
```

```
    p = strtok(NULL, " \n");  
}  
printf("%s\n", line);  
  
}  
printf("-----\n");  
getch();  
}
```